

KNIME WORKFLOW CREATION – Activity

Problem Title: Student Academic Performance Analysis Using KNIME

An educational institution wants to analyze student academic performance using KNIME Analytics Platform. Create a complete KNIME workflow using an Excel dataset containing at least 30 student records with the following attributes:

Student_ID, Student_Name, Department, Gender, Attendance, Internal_Mark, Assignment_Mark, Model_Mark, Final_Mark, Study_Hours, Backlogs

Tasks

Create and execute a KNIME workflow to perform the following:

1. Data Import – 10 Marks
Import the Excel dataset using Excel Reader and verify the data and data types.
2. Data Preprocessing – 10 Marks
Identify and handle missing values using Missing Value node.
3. Column Selection – 5 Marks
Select the relevant columns using Column Filter.
4. Attendance Filtering – 10 Marks
Identify students having Attendance $\geq 75\%$ using Rule-based Row Filter.
5. Total Mark Calculation – 10 Marks
Create a new column Total_Mark using Math Formula/Column Expressions by calculating the total of Internal, Assignment, Model and Final marks.
6. Average Mark Calculation – 5 Marks
Create Average_Mark by calculating the average of the four examination components.
7. Performance Classification – 10 Marks
Using Rule Engine, classify students as:
Excellent (≥ 80), Good (70–79), Average (60–69), Poor (< 60).
8. At-Risk Student Identification – 10 Marks
Using Rule Engine, classify a student as “At Risk” if Attendance < 75 OR Average_Mark < 50 OR Backlogs > 2 ; otherwise classify as “Normal”.
9. Department-wise Analysis – 10 Marks
Using GroupBy, calculate for each department:
Student Count, Average Attendance, Average Mark, Average Study Hours and Total Backlogs.
10. Sorting and Visualization – 10 Marks
Use Sorter to arrange students according to Average_Mark in descending order. Create at least two visualizations showing department-wise average marks and performance-category distribution using suitable chart nodes.
11. Final Output – 10 Marks
Display the final results using Table View and identify:
 - Top-performing students

- Poor-performing students
- At-risk students
- Department with highest average mark
- Department with highest total backlogs

Minimum Required Nodes

Excel Reader → Missing Value → Column Filter → Rule-based Row Filter → Math Formula/Column Expressions → Rule Engine → Sorter → GroupBy → Chart → Table View

Total: 100 Marks

Instruction: Must independently configure, connect, execute and demonstrate the complete KNIME workflow and briefly interpret the obtained results.

ANSWERS

STEP-1:

The screenshot shows the KNIME Analytics Platform interface. The workflow is titled "Student Academic Performance Analysis Using KNIME". The "Excel Reader" node is connected to the workflow. The "Excel Reader" node's configuration panel is open, showing the file path "C:\Users\hp\Downloads\Student_Performance.xlsx". The "Data Area" section is set to "Whole sheet". The "Table" view is selected, showing the first 5 rows of the data table.

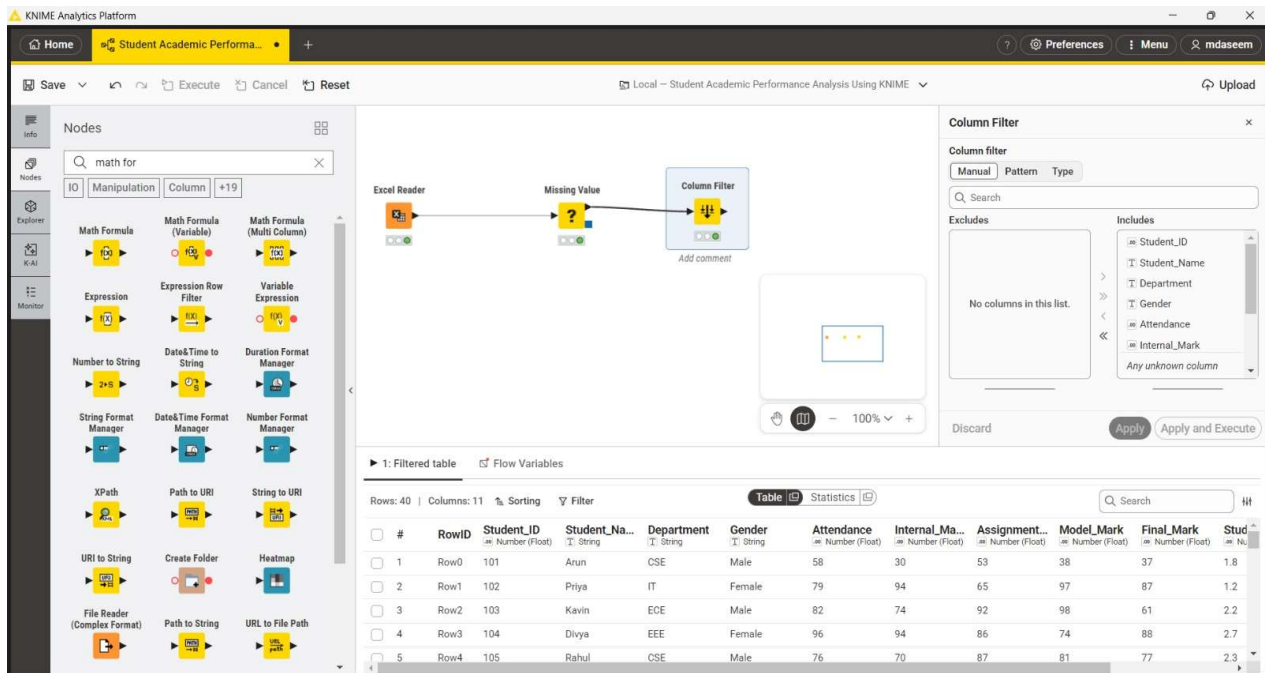
#	RowID	Student_ID	Student_Na...	Department	Gender	Attendance	Internal_Ma...	Assignment...	Model_Mark	Final_Mark	Stud
1	Row0	101	Arun	CSE	Male	58	30	53	38	37	1.8
2	Row1	102	Priya	IT	Female	79	94	65	97	87	1.2
3	Row2	103	Kavin	ECE	Male	82	74	92	98	61	2.2
4	Row3	104	Diviya	EEE	Female	96	94	86	74	88	2.7
5	Row4	105	Rahul	CSE	Male	76	70	87	81	77	2.3

STEP-2:

The screenshot shows the KNIME Analytics Platform interface. The workflow is titled "Student Academic Performance Analysis Using KNIME". The "Excel Reader" node is connected to the workflow. The "Missing Value" node is connected to the "Excel Reader" node. The "Missing Value" node's configuration panel is open, showing the "Treatment by Data Type" section. The "Treatment" dropdown is set to "Median". The "Table" view is selected, showing the first 5 rows of the data table.

#	RowID	Student_ID	Student_Na...	Department	Gender	Attenda...	Internal_Ma...	Assignment...	Model_Mark	Final_Mark	Study
33	Row32	133	Karthik	CSE	Male	93	68	96	96	90	5.7
34	Row33	134	Shalini	IT	Female	89	72	66	66	87	3.5
35	Row34	135	Sathish	ECE	Male	90	63	66	63	85	3
36	Row35	136	Ramya	EEE	Female	58	37	36	36	47	1.8

STEP-3:



KNIME Analytics Platform

Home Student Academic Performance... +

Save Execute Cancel Reset

Local - Student Academic Performance Analysis Using KNIME

Upload

Nodes

Search: math for

Math Formula Math Formula (Variable) Math Formula (Multi Column)

Expression Expression Row Filter Variable Expression

Number to String Date&Time to String Duration Format Manager

String Format Manager Date&Time Format Manager Number Format Manager

XPath Path to URI String to URI

URI to String Create Folder Heatmap

File Reader (Complex Format) Path to String URL to File Path

Column Filter

Column filter

Manual Pattern Type

Search

Excludes

Includes

Student_ID

Student_Name

Department

Gender

Attendance

Internal_Mark

Any unknown column

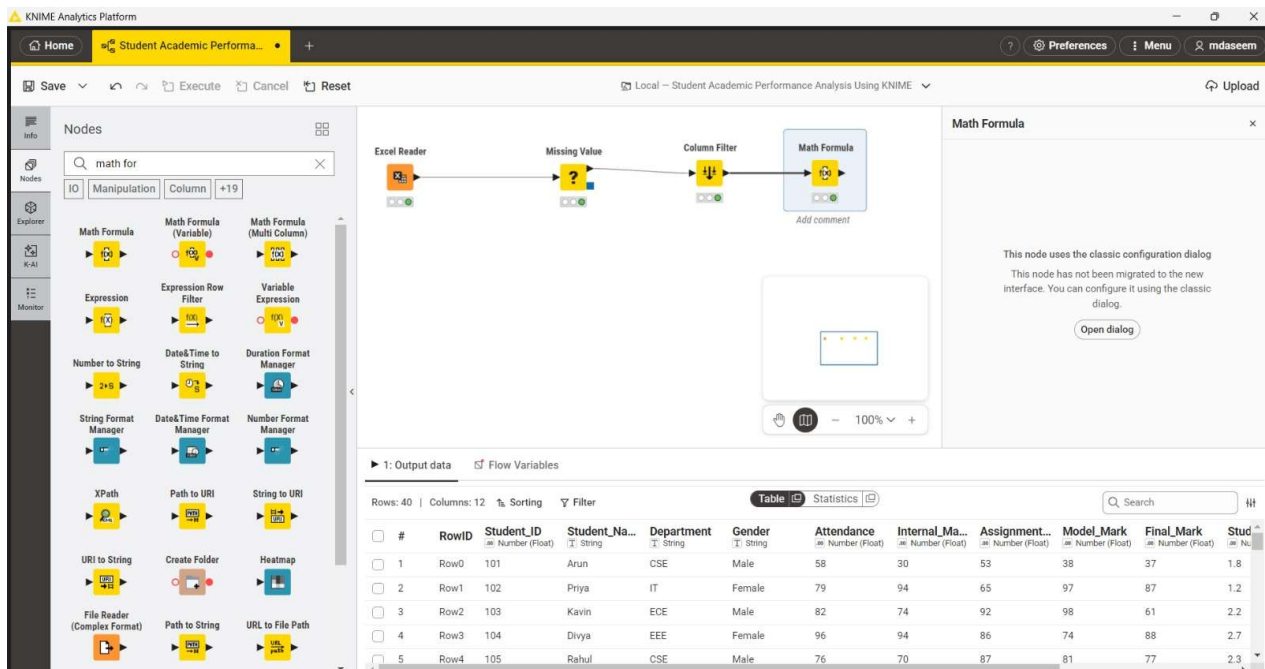
Discard Apply Apply and Execute

1: Filtered table

Flow Variables

#	RowID	Student_ID	Student_Na...	Department	Gender	Attendance	Internal_Ma...	Assignment...	Model_Mark	Final_Mark	Stud...
1	Row0	101	Arun	CSE	Male	58	30	53	38	37	1.8
2	Row1	102	Priya	IT	Female	79	94	65	97	87	1.2
3	Row2	103	Kavin	ECE	Male	82	74	92	98	61	2.2
4	Row3	104	Divya	EEE	Female	96	94	86	74	88	2.7
5	Row4	105	Rahul	CSE	Male	76	70	87	81	77	2.3

STEP-4:



KNIME Analytics Platform

Home Student Academic Performance... +

Save Execute Cancel Reset

Local - Student Academic Performance Analysis Using KNIME

Upload

Nodes

Search: math for

Math Formula Math Formula (Variable) Math Formula (Multi Column)

Expression Expression Row Filter Variable Expression

Number to String Date&Time to String Duration Format Manager

String Format Manager Date&Time Format Manager Number Format Manager

XPath Path to URI String to URI

URI to String Create Folder Heatmap

File Reader (Complex Format) Path to String URL to File Path

Math Formula

This node uses the classic configuration dialog

This node has not been migrated to the new interface. You can configure it using the classic dialog.

Open dialog

1: Output data

Flow Variables

#	RowID	Student_ID	Student_Na...	Department	Gender	Attendance	Internal_Ma...	Assignment...	Model_Mark	Final_Mark	Stud...
1	Row0	101	Arun	CSE	Male	58	30	53	38	37	1.8
2	Row1	102	Priya	IT	Female	79	94	65	97	87	1.2
3	Row2	103	Kavin	ECE	Male	82	74	92	98	61	2.2
4	Row3	104	Divya	EEE	Female	96	94	86	74	88	2.7
5	Row4	105	Rahul	CSE	Male	76	70	87	81	77	2.3

STEP-5:

The KNIME Analytics Platform interface displays a workflow for 'Student Academic Performance Analysis Using KNIME'. The workflow consists of the following nodes: Reader, Missing Value, Column Filter, Math Formula, and Row Filter. The Row Filter node is currently selected, and its configuration panel is open on the right. The configuration panel shows a single criterion: Filter column 'Student_ID' with the operator 'Greater than or equal' and the value '75'. The output section shows 'Column domains' and buttons for 'Discard', 'Apply', and 'Apply and Execute'.

Row Filter Configuration:

- Criterion 1: Filter column: Student_ID, Operator: Greater than or equal, Value: 75
- Output: Column domains
- Buttons: Discard, Apply, Apply and Execute

Table View (1: Included Rows):

#	RowID	Student_ID	Student_Na...	Department	Gender	Attendance	Internal_Ma...	Assignment...	Model_Mark	Final_Mark	Stud
1	Row0	101	Arun	CSE	Male	58	30	53	38	37	1.8
2	Row1	102	Priya	IT	Female	79	94	65	97	87	1.2
3	Row2	103	Kavin	ECE	Male	82	74	92	98	61	2.2
4	Row3	104	Divya	EEE	Female	96	94	86	74	88	2.7
5	Row4	105	Rahul	CSE	Male	76	70	87	81	77	2.3

STEP-6:

The KNIME Analytics Platform interface displays the same workflow as in Step 5, but with the Sorter node added at the end. The Sorter node is selected, and its configuration panel is open on the right. The configuration panel shows a single criterion: Column 'Student_ID' with the order 'Ascending'. The output section shows 'Show advanced settings' and buttons for 'Discard', 'Apply', and 'Apply and Execute'.

Sorter Configuration:

- Criterion 1: Column: Student_ID, Order: Ascending
- Buttons: Discard, Apply, Apply and Execute

Table View (1: Sorted Table):

#	RowID	Student_ID	Student_Na...	Department	Gender	Attendance	Internal_Ma...	Assignment...	Model_Mark	Final_Mark	Stud
1	Row0	101	Arun	CSE	Male	58	30	53	38	37	1.8
2	Row1	102	Priya	IT	Female	79	94	65	97	87	1.2
3	Row2	103	Kavin	ECE	Male	82	74	92	98	61	2.2
4	Row3	104	Divya	EEE	Female	96	94	86	74	88	2.7
5	Row4	105	Rahul	CSE	Male	76	70	87	81	77	2.3

STEP-7:

KNIME Analytics Platform

Local — Student Academic Performance Analysis Using KNIME

Nodes

GroupBy

GroupBy configuration dialog:

This node uses the classic configuration dialog. This node has not been migrated to the new interface. You can configure it using the classic dialog.

Open dialog

► 1: Group table

Rows: 29 | % Sorting | Filter

#	RowID	Final_Mark (as: Number (Float))
1	Row0	36
2	Row1	37
3	Row2	44
4	Row3	47
5	Row4	53

STEP-8:





STEP-9:

FINAL OUTPUT:

Interactive View: Table View

Table View

Rows: 40 | Columns: 15

RowID	Student_ID	Student_...	Departm...	Gender	Attendan...	Internal_...	Assignm...	Model_M...	Final_Mark	Study_Ho...	Backlogs	Total_Mark	Average_...	Performa...	
	Number (...)	T: String	T: String	T: String	Number (...)	Number (...)	Number (...)	Number (...)	Number (...)	Number (...)	Number (...)	Number (...)	Number (...)	T: String	
Row0	101	Arun	CSE	Male	58	30	53	38	37	1.8	3	158	39.5	Poor	At F
Row1	102	Priya	IT	Female	79	94	65	97	87	1.2	0	343	85.75	Excellent	Nor
Row2	103	Kavin	ECE	Male	82	74	92	98	61	2.2	2	325	81.25	Excellent	Nor
Row3	104	Divya	EEE	Female	96	94	86	74	88	2.7	2	342	85.5	Excellent	Nor
Row4	105	Rahul	CSE	Male	76	70	87	81	77	2.3	0	315	78.75	Good	Nor
Row5	106	Sneha	IT	Female	86	66	65	84	66	3.7	1	281	70.25	Good	Nor
Row6	107	Vijay	ECE	Male	84	63	56	69	72	6.8	1	260	65	Average	Nor
Row7	108	Meena	EEE	Female	67	32	47	39	56	4.7	5	174	43.5	Poor	At F
Row8	109	Ajay	CSE	Male	87	96	72	64	62	2.4	2	294	73.5	Good	Nor
Row9	110	Nisha	IT	Female	85	65	74	66	84	3.7	1	289	72.25	Good	Nor
Row10	111	Hari	ECE	Male	87	70	83	82	73	2.6	2	308	77	Good	Nor
Row11	112	Keerthi	EEE	Female	97	64	98	70	94	2.5	2	326	81.5	Excellent	Nor
Row12	113	Surya	CSE	Male	82	67	72	75	72	5.1	1	286	71.5	Good	Nor
Row13	114	Anitha	IT	Female	77	74	62	80	85	1.4	1	301	75.25	Good	Nor
Row14	115	Gokul	ECE	Male	73	58	52	40	36	4	5	186	46.5	Poor	At F
Row15	116	Lavanya	EEE	Female	96	89	69	76	68	5.5	0	302	75.5	Good	Nor

Interactive View: Table View

Table View

Rows: 40 | Columns: 15

D	Student_ID = Number (-)	Student_... T: String	Departm... T: String	Gender T: String	Attendan... = Number (-)	Internal_... = Number (-)	Assignm... = Number (-)	Model_M... = Number (-)	Final_Mark = Number (-)	Study_Ho... = Number (-)	Backlogs = Number (-)	Total_Mark = Number (-)	Average... = Number (-)	Performa... T: String	Risk statu... T: String
1	101	Arun	CSE	Male	58	30	53	38	37	1.8	3	158	39.5	Poor	At Risk
	102	Priya	IT	Female	79	94	65	97	87	1.2	0	343	85.75	Excellent	Normal
2	103	Kavin	ECE	Male	82	74	92	98	61	2.2	2	325	81.25	Excellent	Normal
1	104	Divya	EEE	Female	96	94	86	74	88	2.7	2	342	85.5	Excellent	Normal
1	105	Rahul	CSE	Male	76	70	87	81	77	2.3	0	315	78.75	Good	Normal
3	106	Sneha	IT	Female	86	66	65	84	66	3.7	1	281	70.25	Good	Normal
3	107	Vijay	ECE	Male	84	63	56	69	72	6.8	1	260	65	Average	Normal
7	108	Meena	EEE	Female	67	32	47	39	56	4.7	5	174	43.5	Poor	At Risk
1	109	Ajay	CSE	Male	87	96	72	64	62	2.4	2	294	73.5	Good	Normal
1	110	Nisha	IT	Female	85	65	74	66	84	3.7	1	289	72.25	Good	Normal
0	111	Hari	ECE	Male	87	70	83	82	73	2.6	2	308	77	Good	Normal
1	112	Keerthi	EEE	Female	97	64	98	70	94	2.5	2	326	81.5	Excellent	Normal
2	113	Surya	CSE	Male	82	67	72	75	72	5.1	1	286	71.5	Good	Normal
3	114	Anitha	IT	Female	77	74	62	80	85	1.4	1	301	75.25	Good	Normal
4	115	Gokul	ECE	Male	73	58	52	40	36	4	5	186	46.5	Poor	At Risk
5	116	Lavanya	EEE	Female	96	89	69	76	68	5.5	0	302	75.5	Good	Normal